When does A* save work?

T01: structure, heuristic and the cost of guidance

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Defence

The question

Dijkstra expands every vertex closer than the target.
A* adds a heuristic $h(v)$ and expands fewer vertices.

But guidance is not free:

- one heuristic evaluation per generated vertex;
- larger heap keys / more memory traffic;
- a weak heuristic can save nothing.

Research question

How do graph structure and heuristic strength change

- expanded vertices and
- wall-clock runtime

of A* versus Dijkstra, on unit-cost grids and a real road graph?

The answer must separate **expansions** from **time**.

Data: synthetic controls and a real graph

Grids — 4-neighbour, unit cost, 128–1024 square

- open
- random 20% obstacles
- clustered 30% obstacles
- rooms (walls with doors)
- perfect maze

Largest 4-connected component is used. Blocked cells are isolated vertices.

Real graph: SNAP `roadNet-CA`

- California road network
- 1,957,027 vertices after symmetrisation and largest component
- 2,760,388 undirected edges (5,520,776 CSR arcs)
- unit hop weight: SNAP has no lengths/travel times

One sparse, planar-like road graph; one limitation, not a sample of all real graphs.

experiment	query instances	timed runs
grids (20 graphs)	300	2,700
real graph	30	900
total	330	3,600

What we built: ADTs and complexity

Graph: CSR

`offsets[n+1]`, `targets[m]`, `weights[m]`
memory $O(n+m)$; neighbour iteration $O(\deg u)$

Binary heap

push/pop $O(\log n)$ worst case; top $O(1)$
lazy deletion: no decrease-key; stale entries are detected on pop
ties: `node_id` vs `larger_g`

One search loop

Dijkstra = same loop with $h \equiv 0$
A* = $f = g + h(v)$; stop when target is settled
consistent heuristic $\Rightarrow$ no reopening
scratch arrays reset in $O(1)$ by generation stamp

Heuristics

grids: Manhattan (admissible + consistent)
road: ALT $\max_L |d(L,v) - d(L,t)|$
16 landmarks by farthest-point sampling; main $K = 8$

Correctness first

Tests before any timing

- heap order vs reference priority queue
- hand-computed 5x5 wall grid: distance 8
- hand-computed weighted graph vs Floyd–Warshall
- all five grid families vs independent BFS
- ALT admissibility and consistency for every free vertex
- path validation: contiguous edges, weights sum to distance

Oracle and audit

Every benchmark query is checked against BFS distance.

3,600 / 3,600 runs: `status=ok`, distance matches oracle, zero timeouts.

Expansions deterministic across repeats; median time CV $\approx$ 7–10%.

counter	definition (short)
expanded	popped, not stale, not closed; target included
pushed	source + every successful distance decrease
stale_popped	old heap entry or already-closed vertex
time_ns	search call only; inputs outside timer

Frozen protocol

- paired Dijkstra/A* on exactly the same query
- 15 grid queries/graph, 30 real queries
- 3 grid repeats, 5 real repeats
- one untimed warm-up per treatment
- report median per query; keep raw repeats
- bootstrap 95% CI + paired Wilcoxon
- 30 s cap per search; 4 GiB process cap
- one CPU via `taskset`

Query bands

BFS distance from a random source:

- short: 8–25% of eccentricity
- medium: 40–60%
- long: 80–97%

Selection uses only BFS, never Dijkstra/A*.

Freeze

Predictions in `HYPOTHESES.md`; protocol fixed before the main run in `PROTOCOL.md`.

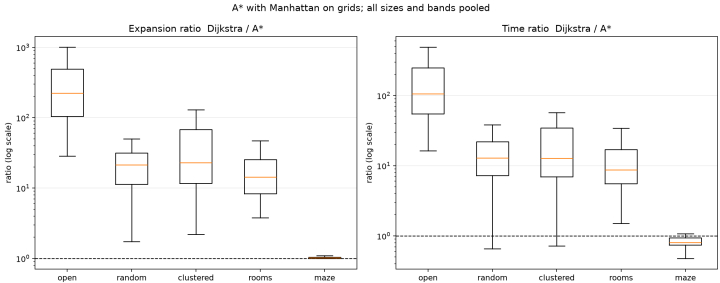
Predictions before the main run

	prediction	verdict
H1	open grid: A* saves many expansions and time	confirmed
H2	random obstacles: still saves, less than open	confirmed
H3	clustered/rooms: weaker than open; rooms weaker than random	confirmed
H4	maze: no expansion saving; A* slower	confirmed
H5	long queries benefit more than short	partly rejected
H6	ALT strong at K=8; K=1 may not help; diminishing at 16	partly rejected
H7	`larger_g` reduces median expansions on open	rejected as median claim

The protocol explicitly allowed negative and null results.

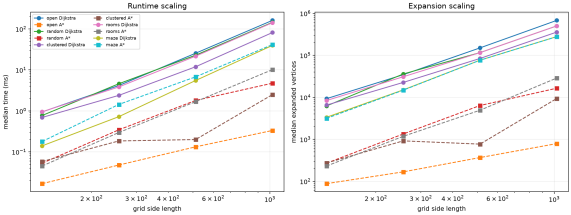
Main grid result

family	time ratio (median)	expansion ratio (median)	time saving (median)	A* faster queries
open	**105.7x**	**222.4x**	99.1%	100%
random 20%	12.8x	21.4x	92.2%	98%
clustered 30%	12.8x	22.9x	92.2%	98%
rooms	8.7x	14.3x	88.5%	100%
maze	**0.81x**	**1.02x**	**–24.0%**	17%



Ratio = Dijkstra / A*; values above 1 mean A* wins.

Structure matters more than distance



Open grid: A* follows one corridor; the advantage grows with size (390x time, 696x expansions at 1024).

Random/clustered: local detours cost some guidance power; still 8–13x time.

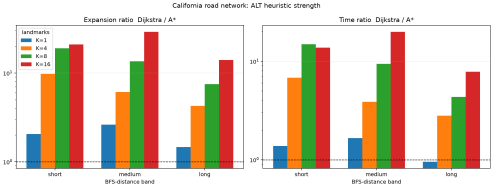
Rooms: long forced detours; weakest positive family (8.7x).

Maze: Manhattan cannot see the forced turns. A* is **24% slower**; expansions are essentially unchanged.

Real road network: ALT makes A* worthwhile

ALT	time ratio	exp. ratio	time saving	faster
K=1	1.29x	1.96x	22.5%	70%
K=4	4.75x	7.26x	78.9%	100%
K=8	**7.62x**	**11.54x**	86.8%	100%
K=16	**13.52x**	**22.41x**	92.6%	100%

K=1 time difference not significant (p=0.13); K=4/8/16 p<1e-8.
K=16 is 21% faster than K=8 on paired queries.



More landmarks keep tightening the bound; no plateau at 8 on this graph.

Why time saving is smaller than expansion saving

family	Dijkstra ns/exp	A* ns/exp	ratio
open	142	288	2.03x
random	157	254	1.62x
clustered	127	245	1.93x
rooms	146	270	1.85x
maze	65	79	1.21x

Real graph: Dijkstra 235 ns/exp; ALT K=8 373 ns (1.59x); K=16 394 ns (1.68x).

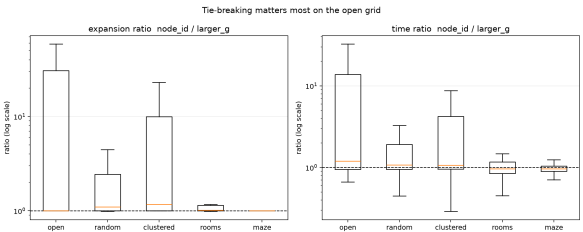


The rule: time is saved only when the expansion reduction pays for the more expensive expansion.

Tie-breaking: a tail-risk decision

family	median ratio	node_id worse queries	worst ratio
open	1.00	45%	285.7x
random	1.10	63%	51.5x
clustered	1.17	68%	92.6x
rooms	1.02	58%	17.8x
maze	1.00	2%	1.01x

Ratio = node_id expansions / larger_g expansions.
On open, `larger_g` never expanded more; it removes the catastrophic tail.

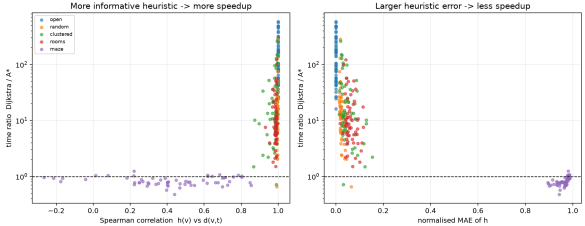


What makes a heuristic informative?

family	h corr.	norm. MAE	time ratio
open	1.00	0.00	105.7x
random	1.00	0.02	12.8x
clustered	0.99	0.04	12.8x
rooms	0.99	0.06	8.7x
maze	0.42	0.97	0.81x

Pooled Spearman: $+0.78$ for h-distance correlation, -0.81 for normalised MAE. A one-feature model has $R^2 = 0.55$.

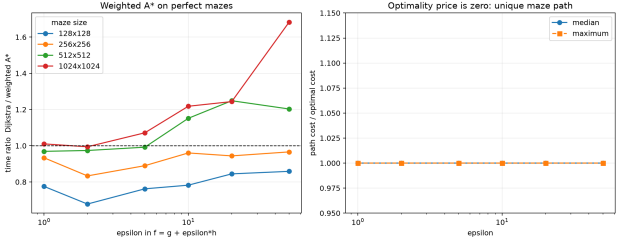
Admissibility is not enough: the heuristic must be *informative* about the actual remaining distance.



Can weighted A* recover the maze?

ϵ	time ratio	cost ratio	1024^2
1	0.96	1.00	1.01x
10	1.10	1.00	1.22x
20	1.15	1.00	1.24x
50	1.18	1.00	1.68x

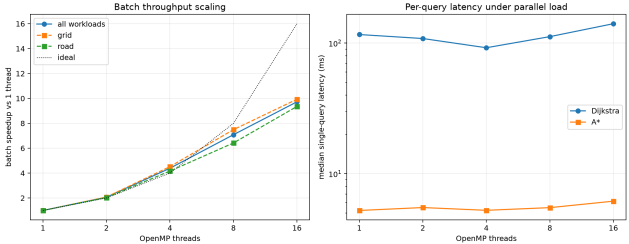
$f = g + \epsilon h$, with reopening. A perfect maze has a unique simple path, so every returned path has the same cost. At 1024^2 and $\epsilon = 50$ the median speedup is 1.68x; short queries reach 3.43x. At 128^2 the overhead still loses.



Does A* still win with 16 threads?

threads	batch	speedup	efficiency
1	44.39 s	1.00x	1.00
2	21.48 s	2.07x	1.03
4	10.11 s	4.39x	1.10
8	6.27 s	7.08x	0.89
16	4.56 s	**9.73x**	0.61

Five 1024^2 grids + 30 road queries, 3 repeats. Each task runs Dijkstra and A* on the same query; thread-local scratch and heuristic. A* advantage remains: median Dijkstra/A* ratio 14.8x at one thread, 17.9x at 16. Maze stays below parity.



Throughput gain, not single-query speedup.

What the data rejected

- **H5:** long queries do not always benefit more. Open grid: long 156x vs short 83x time; random grid: long 11.5x vs short 14.4x.
- **H6 (diminishing returns):** K=16 is still better than K=8 on paired real queries (21% time, 25% expansions, p<1e-4).
- **H7 (median tie-break):** open median is exactly 1.00, although 45% of queries are worse with `node_id`, up to 286x.
- **The headline negative result:** in the maze, A* is 24% slower even though its distances are correct.

Interpretation

A null or negative result is still a result: guidance only pays when it changes the explored region, not merely when it is admissible.

Threats to validity

- one real graph; road network, not a sample of all real graphs
- unit hop weights; no travel times in SNAP
- symmetrised roads; one-way streets not modelled
- synthetic grid families are controls, not real maps
- one machine, one heap implementation

- medians and IQRs, bootstrap CIs, paired Wilcoxon
- within-query time CV 7–10%
- all 3,600 runs match BFS; no timeouts
- raw CSV and seeds are kept
- protocol frozen before the main run

The constants are implementation- and machine-specific; the direction and the mechanism are the transferable result.

Final answer

When does A* save work?

When the heuristic is informative about the *actual* remaining path, and the reduction in expanded vertices is large enough to pay for the more expensive expansion.

- **Open grid:** yes, spectacularly (106x time, 222x expansions).
- **Random/clustered/rooms:** yes, but 9–13x time, not 100x.
- **Maze:** no; 24% slower, because Manhattan cannot see walls.
- **Road network:** yes; ALT K=8 7.6x, K=16 13.5x.
- **Practical rule:** measure expansions and cost per expansion; do not assume the heuristic is free.

Reproduce

Build and test

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j
./build/test_core
```

Full frozen run

```
./scripts/run_experiments.sh
```

Artifacts

`HYPOTHESES.md`, `PROTOCOL.md`, `RESULTS.md`,
`results/grid.csv`, `results/real.csv`, `results/analysis.md`, `figures/`

Questions?